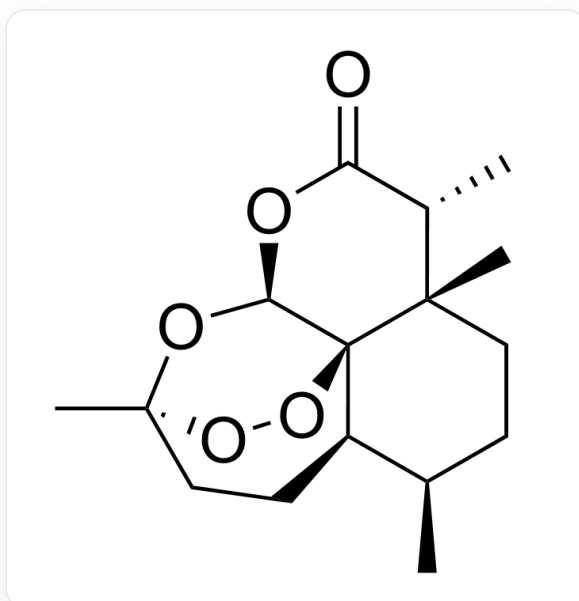


Question

Some laboratories use the following method to determine the content of substance **A** in a certain plant. The structure of **A** is shown below:



C[C@@H]1CC[C@@]2(C)[C@@H](C)C(=O)O[C@H]3[C@]42[C@@H]1CC[C@@](C)(O3)OO4

Precisely weigh 0.2885 g of potassium dichromate dried at 130°C for two hours, dissolve it in water, and dilute to 100 mL. Transfer 20.00 mL, add 1 g of potassium iodide and 5 mL of 1+5 sulfuric acid, and let it stand in the dark for 5 min. Using starch as the indicator, titrate with sodium thiosulfate solution until the endpoint is reached, consuming 26.24 mL. Then weigh 24.9034 g of the dried plant powder into a dry 250 mL round-bottom flask, add 200 mL of n-pentane, install a Soxhlet extractor, and reflux for 2 h. Filter under suction, dilute with n-pentane to 250 mL. Transfer 25.00 mL of the n-pentane solution, evaporate the n-pentane on a water bath, add 10 mL of anhydrous ethanol, 1 mL of glacial acetic acid, 20 mL of 5% potassium iodide, and 5 mL of 1+1 hydrochloric acid, heat on a water bath at 70°C for 30 minutes, and titrate with sodium thiosulfate using starch as the indicator until the endpoint is reached, consuming 11.73 mL of the standard sodium thiosulfate titration solution.

Among the following statements, at least one is correct (for statements involving calculations, deviations within 1% are considered correct):

1. There are 8 chiral carbon atoms in the structure of **A**.
2. In the structure of **A**, there is a carbon atom shared by three six-membered rings, and its stereochemical configuration is R.
3. When extracting **A** using a Soxhlet extractor, n-pentane can be replaced with n-hexane.
4. The concentration of the sodium thiosulfate solution is 0.04585 mol/L.
5. The content of **A** is 3.000%.

Among these statements, let the number of incorrect statements be a, and the smallest number among the correct statements be b. Then a and b are respectively

A. 1, 1

B. 1, 2

C. 1, 3

D. 1, 4

E. 1, 5

F. 2, 1

G. 2, 2

H. 2, 3

I. 2, 4

J. 2, 5

K. 3, 1

L. 3, 2

M. 3, 3

N. 3, 4

O. 3, 5

P. 4, 1

Q. 4, 2

R. 4, 3

S. 4, 4

T. 4, 5

Answer

Correct Answer: **L**

Detailed Explanation

According to the structure, **A** actually has 7 chiral carbon atoms; Statement 1 is incorrect.

CHECKPOINT

1 PTS

A actually has 7 chiral carbon atoms

The carbon atom shared by three six-membered rings (one of which is a peroxide six-membered ring, relatively concealed) is the central carbon atom of the structure, with an R stereochemical configuration. Therefore, Statement 2 is correct.

CHECKPOINT

1 PTS

The carbon atom shared by three six-membered rings is the central carbon atom of the structure, with an R stereochemical configuration

A contains a bridged ring structure formed by a peroxide bond, which is thermally unstable and prone to decomposition. Thus, n-hexane with too high a boiling point cannot be used as the extraction solvent, making Statement 3 incorrect.

CHECKPOINT

1 PTS

A is thermally unstable and prone to decomposition, so n-hexane with too high a boiling point cannot be used as the extraction solvent

The analysis process of **A** consists of two steps. First, the sodium thiosulfate solution is standardized using potassium dichromate. According to the stoichiometric relationship, 1 mol $\text{Cr}_2\text{O}_7^{2-}$ produces 3 mol I_2 , and 1 mol I_2 consumes 2 mol $\text{S}_2\text{O}_3^{2-}$. Therefore, $n(\text{Cr}_2\text{O}_7^{2-}) : n(\text{I}_2) = 1 : 6$. Substituting the data yields a sodium thiosulfate solution concentration of 0.04485 mol/L, which differs from the 0.04585 mol/L in Statement 4 by more than 1%. Hence, Statement 4 is incorrect.

CHECKPOINT

1 PTS

The sodium thiosulfate solution concentration is 0.04485 mol/L

The peroxide bond in **A** can undergo a redox reaction with potassium iodide, and the generated iodine is back-titrated with sodium thiosulfate solution. The stoichiometric relationship is 1 mol **A** $\sim$ 1 mol I_2 $\sim$ 2 mol $\text{S}_2\text{O}_3^{2-}$. Substituting the specific data gives a total mass of **A** in the sample as 0.7425 g, with a mass percentage of 2.982%, differing from the 3.000% in Statement 5 by less than 1%. Thus, Statement 5 is correct.

CHECKPOINT

2 PTS

The total mass of **A** in the sample is 0.7425 g, with a mass percentage of 2.982%

Therefore, there are 3 incorrect statements in total, and the correct statements are Statements 2 and 5, making Option L correct.

In fact, **A** is artemisinin.